

SQL REASONING QUICK CHEATSHEET

LAST-MINUTE PLACEMENT EXAM REVISION (1-2 MIN SCAN)



BASIC SYNTAX & DATATYPES



- SELECT commands datats
- FROM employees
- WHERE: INT, VARCHAR, DATE, BOOLEAN
- Table visual

```
create table (
  name, VARCHAR (50,
  dept: INT0)
);
```



STRING & DATE HANDLING



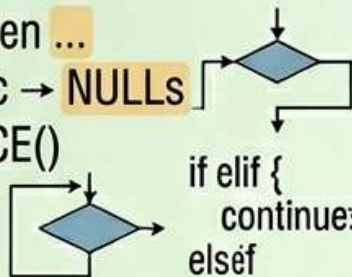
- LENGTH() $s = \text{"example"}$
- UPPER()
- LOWER(), SUBST(start, length)
- GETDATE(), YEAR()
- CONCAT(), DATEPART()



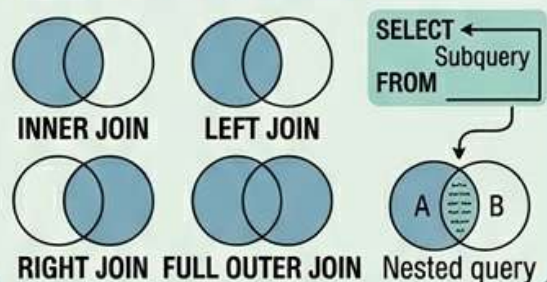
CONTROL FLOW (CASE, COALESCE)



- CASE when ... when it ic → NULLs
- COALESCE()
- conditio- case



JOINS & SUBQUERIES



AGGREGATE FUNCTIONS & AGGREGATION



- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()

id	salary	tomv	dept
1	2	4th	4
2	3	3th	3
3	4	300.	3

GROUP BY HAVING

```
SELECT SUM(salary)
FROM employees
GROUP BY dept;
```



NULL & TRANSACTION HANDLING



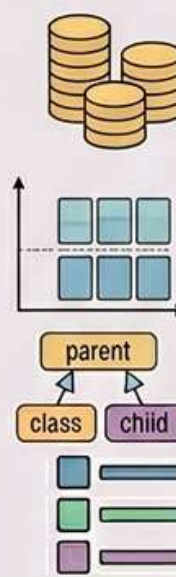
- handling NULL values NULL
- IS NULL, IS NOT NULL
- NVL() (or IFNULL)
- NULLIF()
- transactions BEGIN TRANSACTION COMMIT ROLLBACK



ADVANCED CONCEPTS (WINDOW, CTEs)



- ROW_NUMBER()
- RANK()
- DENSE_RANK()
- LEAD()
- LAG()
- PARTITION BY
- CTE WITH table_name AS (...)



INDEXES, CONSTRAINTS & SET OPERATIONS



- creating indexes for speed
- Constraints
 - PRIMARY KEY
 - FOREIGN KEY
 - UNIQUE
 - NOT NULL
- SET operations
 - UNION
 - UNION ALL
 - INTERSECT
 - EXCEPT

